Topic 2: System-Level Design Considerations

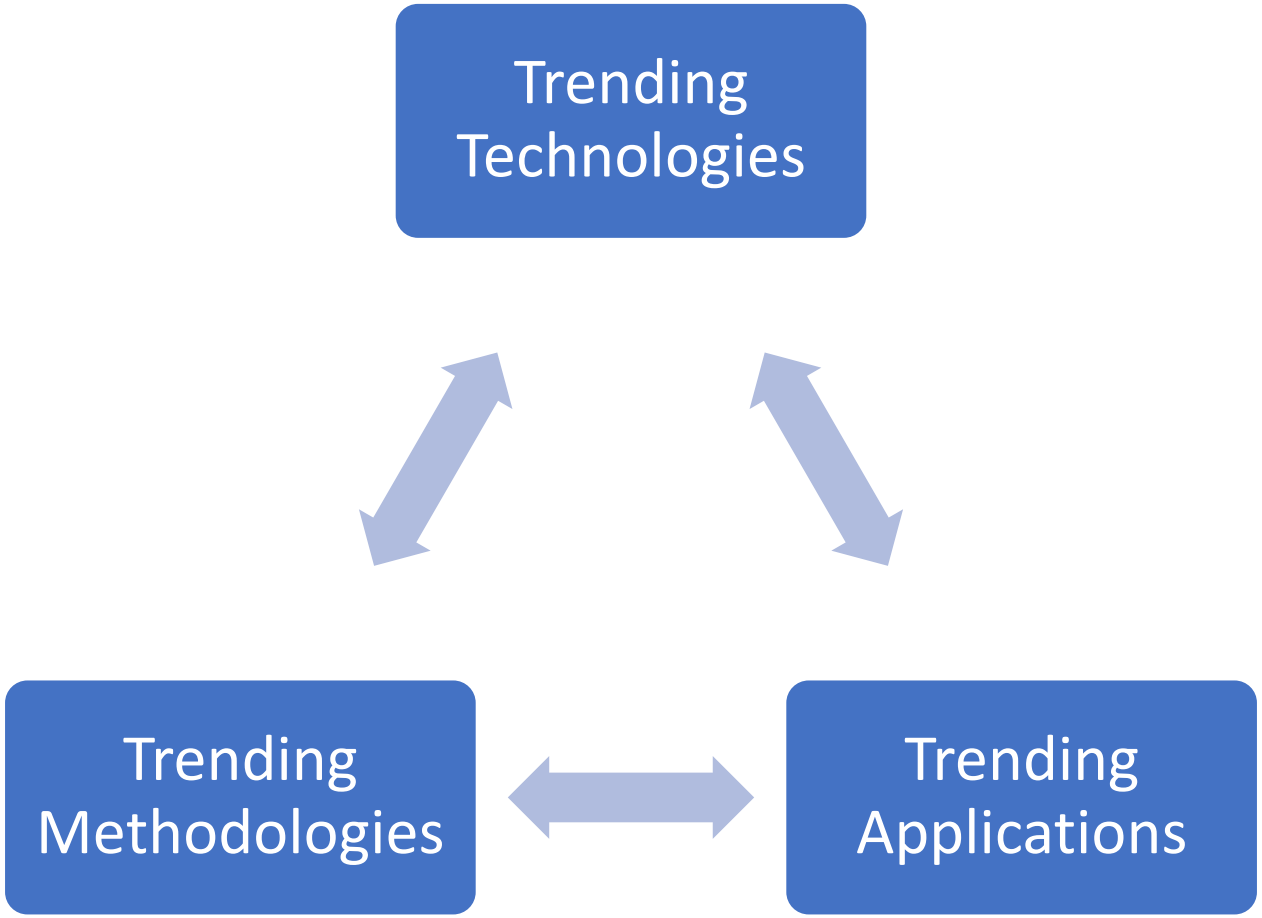
ECE 9433: Agile System-on-Chip Design

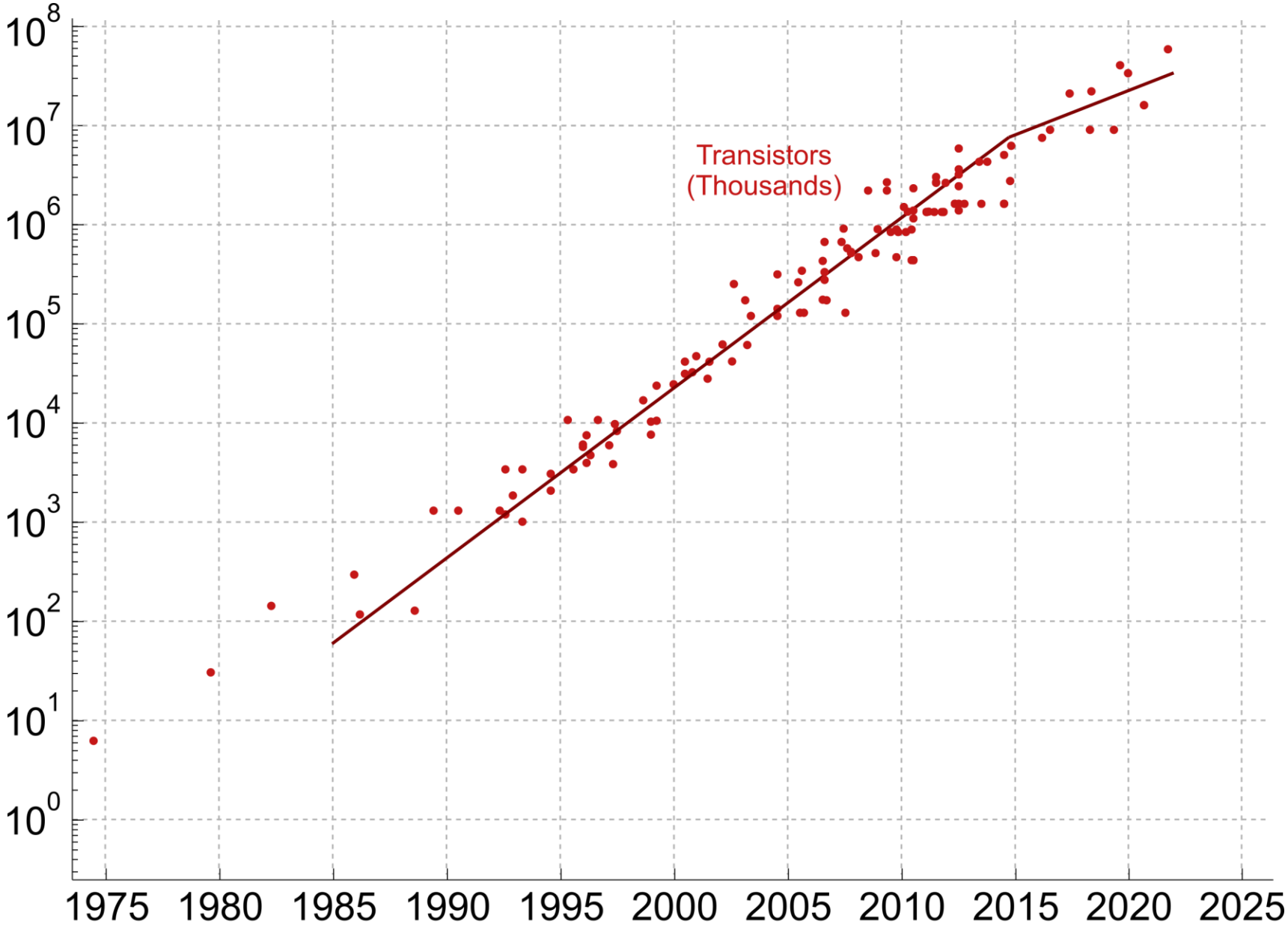
Fall 2024

Prof. Austin Rovinski

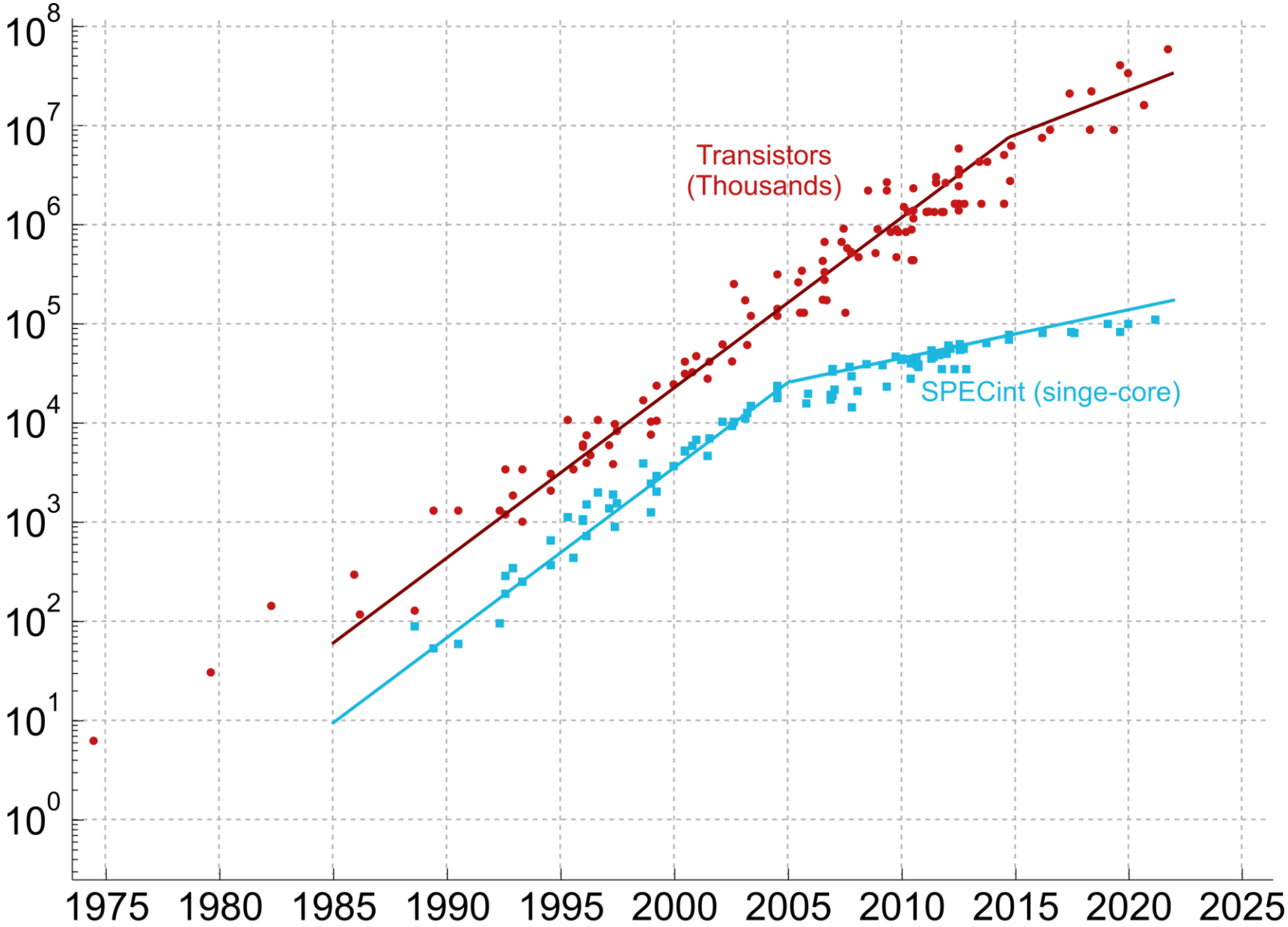
New York University

Modern Semiconductor Trends





C. Batten, M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, K. Rupp & [Y. Shao, IEEE Micro'15] & [C. Leiserson, Science'20]



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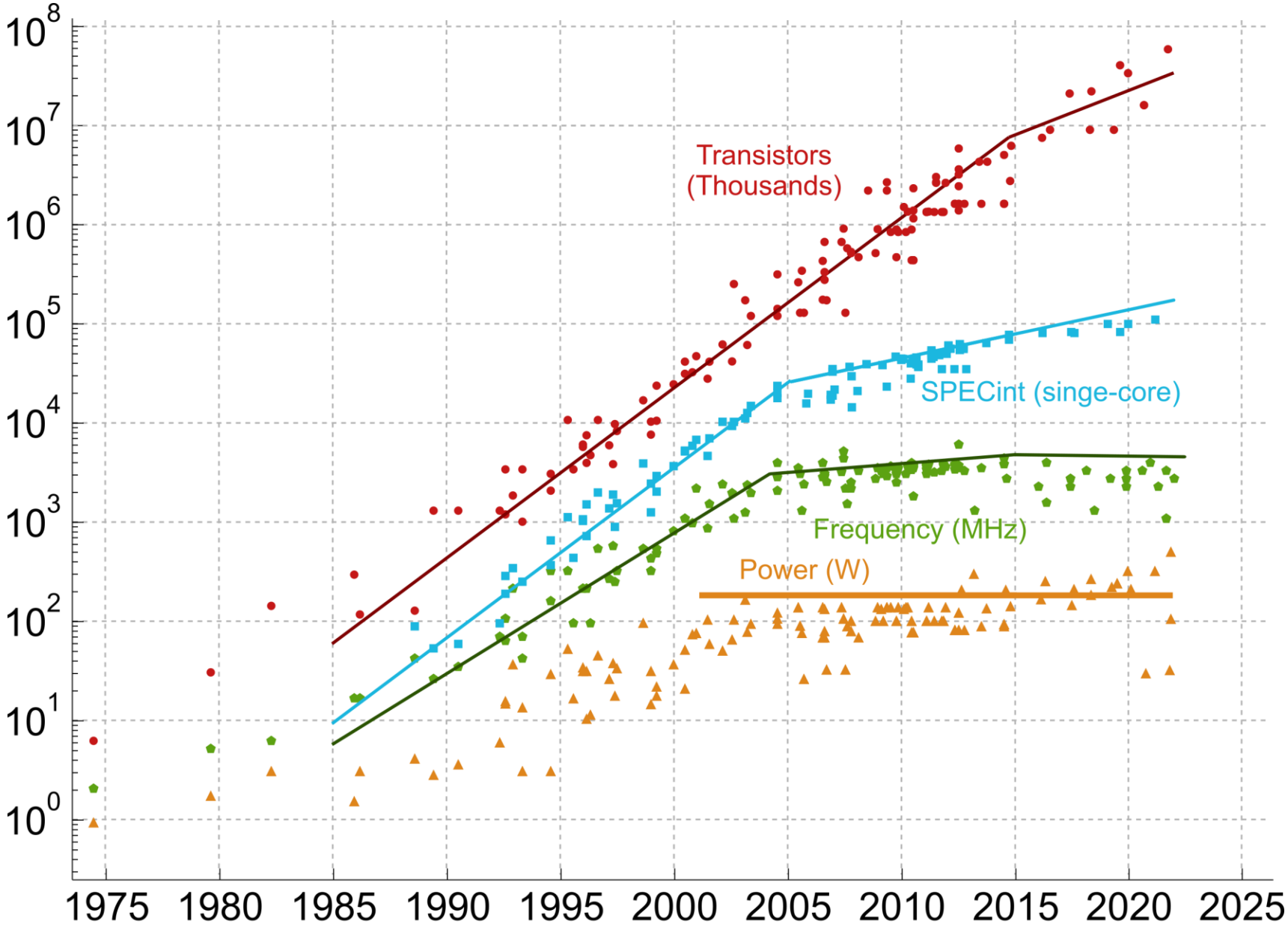

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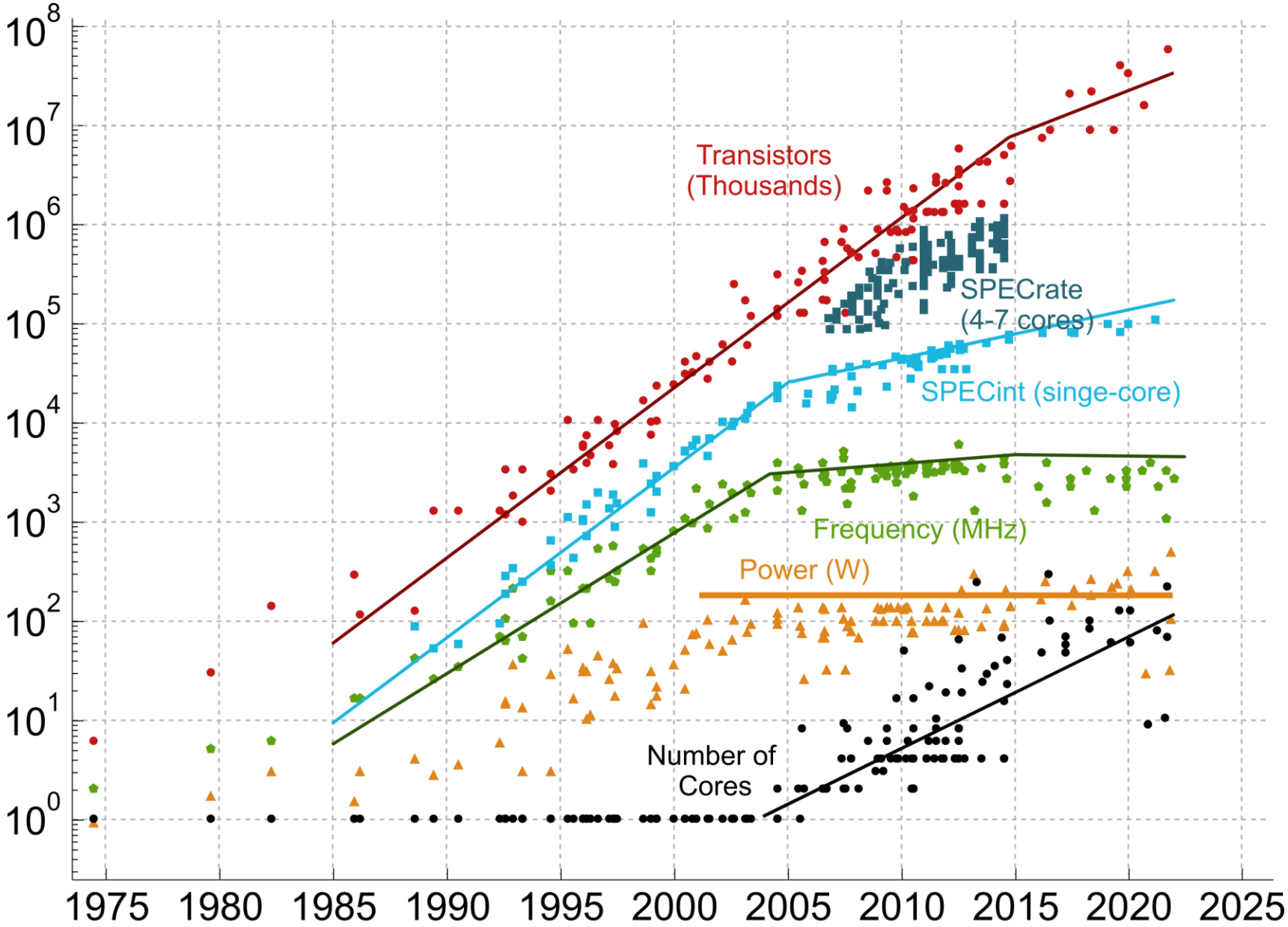

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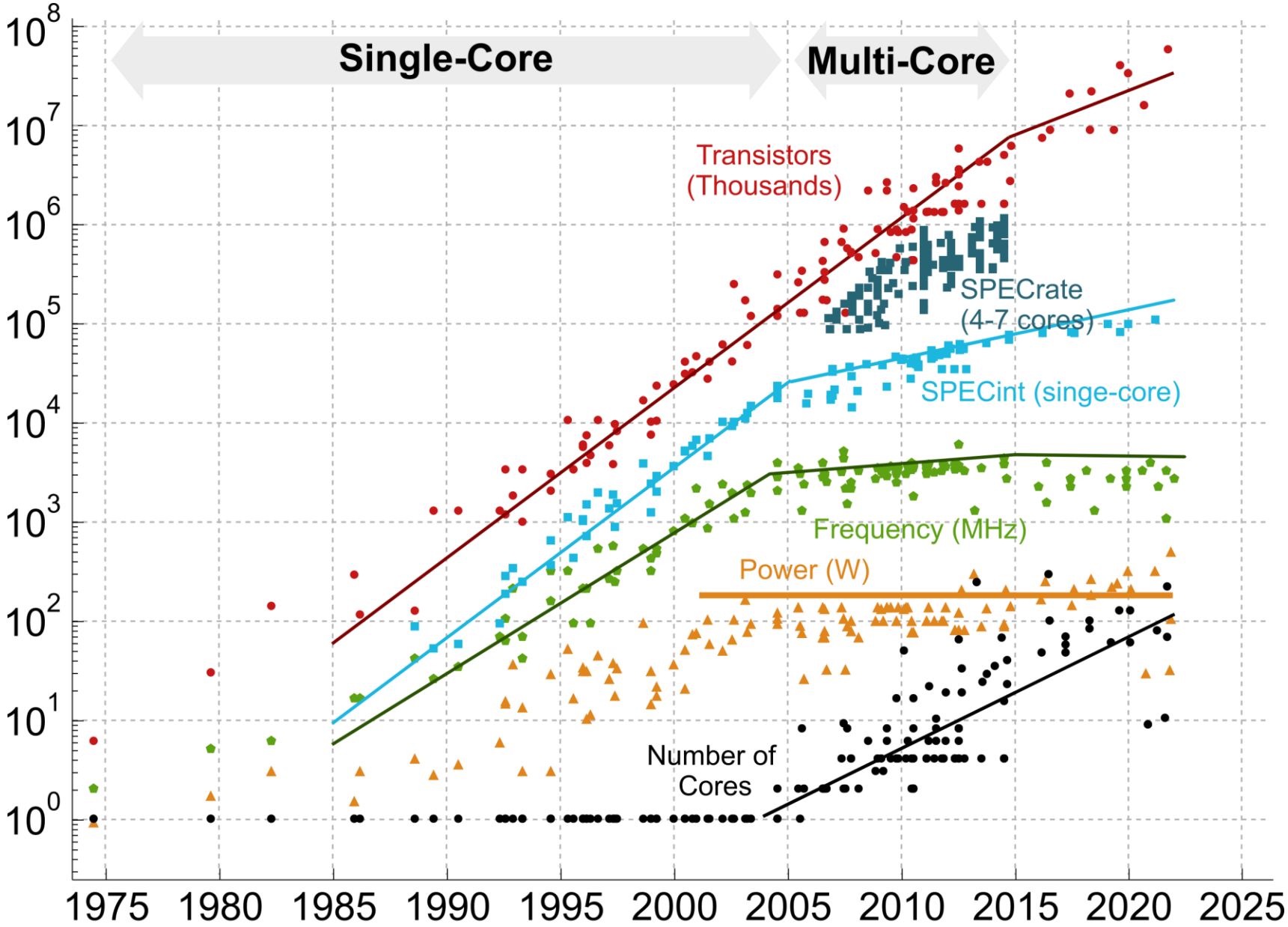

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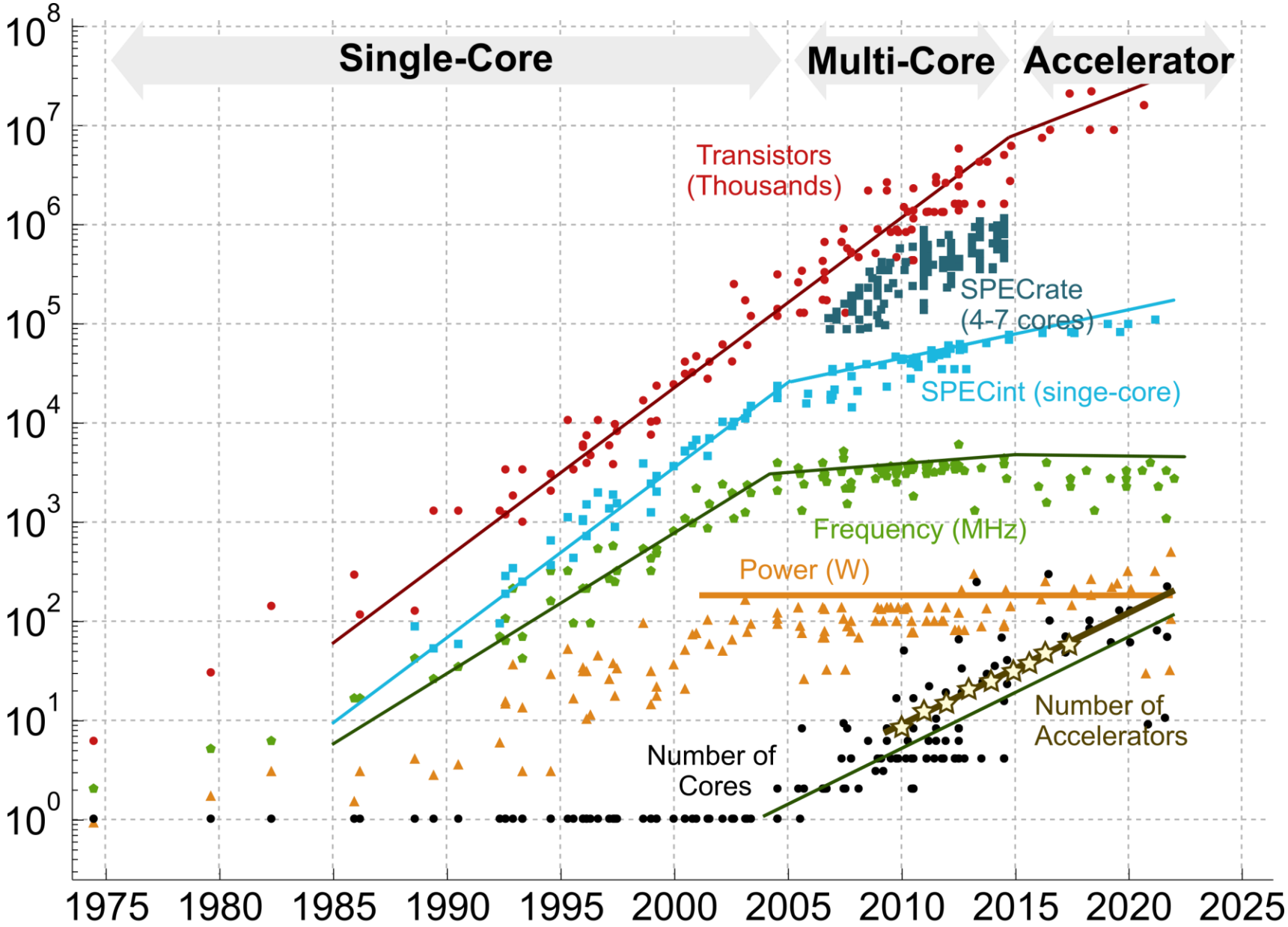

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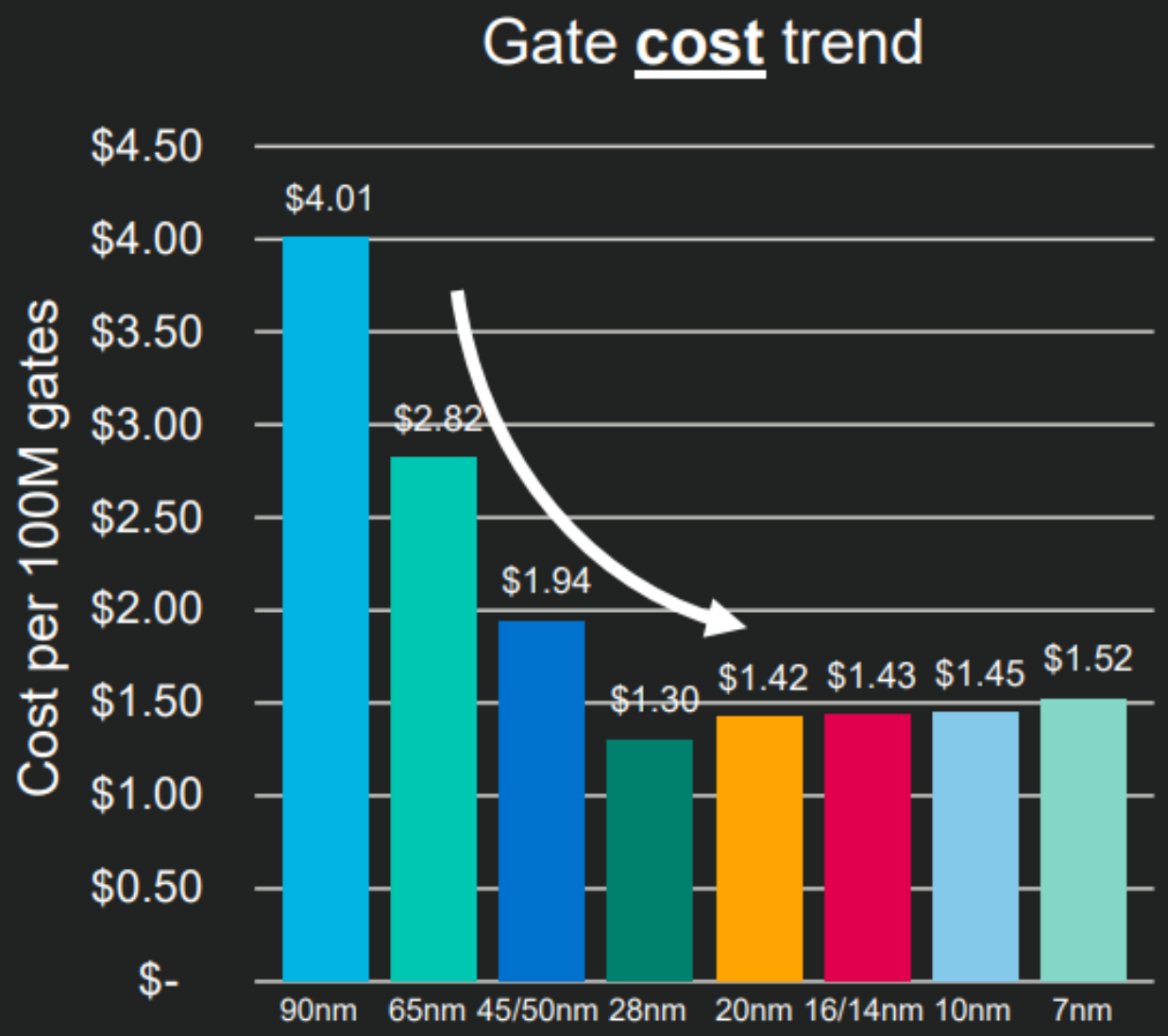
Technology Trends

- Transistor scaling is slowing to a crawl
- Prior general-purpose paradigms no longer present efficiency gains
- Some MOSFET gains are still on the table
  - Planar transistors transitioned to FinFET, soon to be GAAFET
  - Backside power delivery leaves room for substantial density increase
- Scalability through interconnects and packaging are still available

More on these later

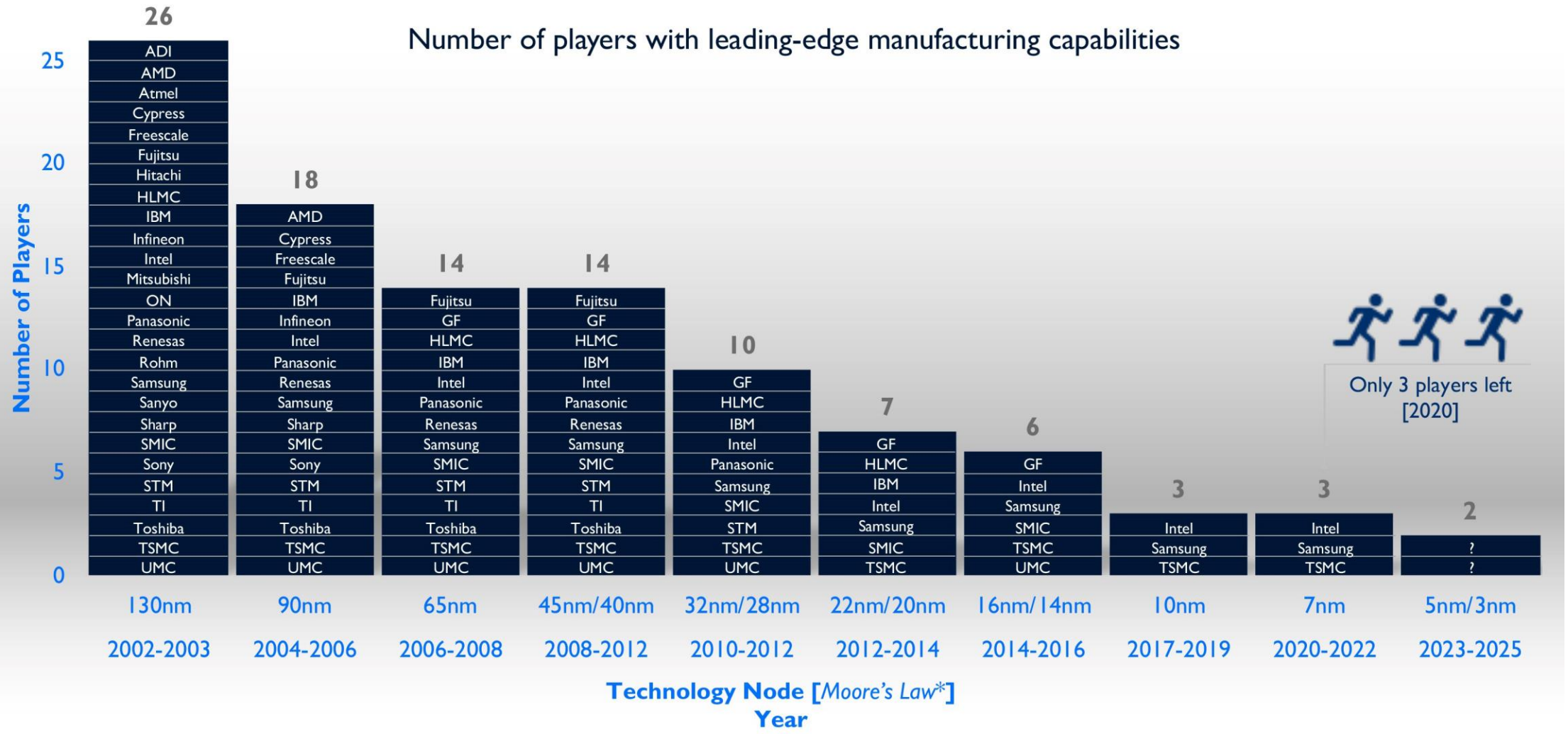
Moore's law slowing down…



Source: Marvell
2020 Investor Day

Semiconductor industry evolution

(Source: High-End Performance Packaging: 3D/2.5D Integration report, Yole Développement, 2020)



* Moore's law states that the number of transistors in an integrated circuit chips doubles every 2 years
Data referenced from Intel and WikiChip

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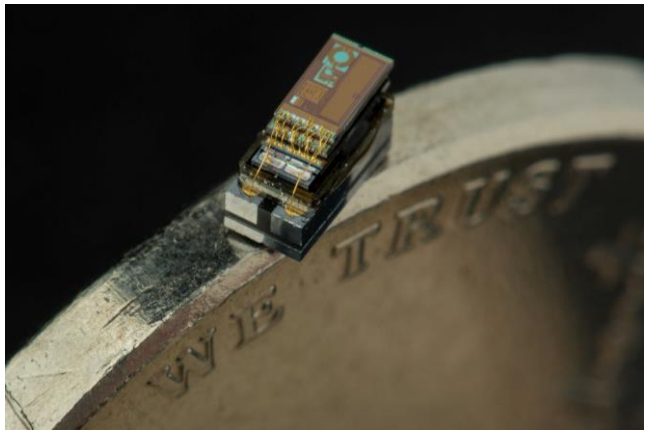
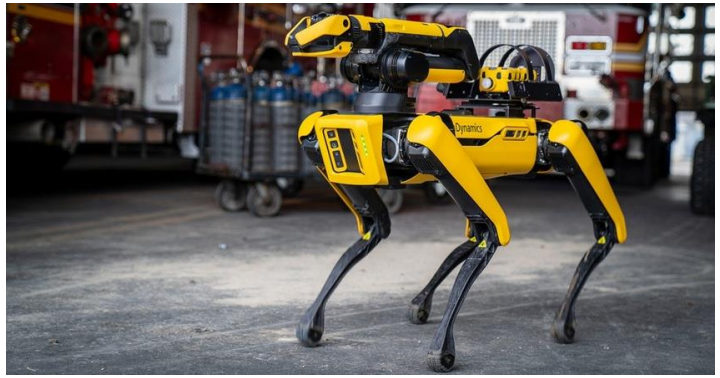
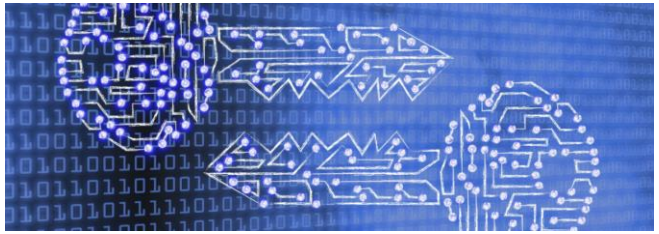
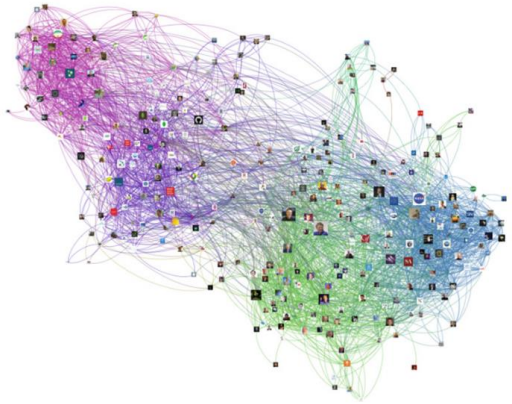
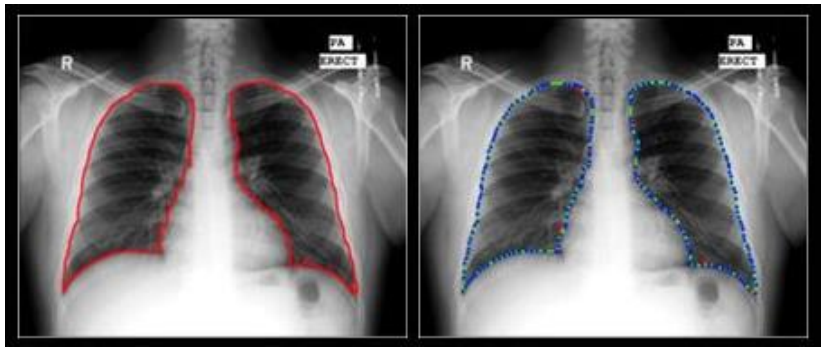
Methodology Trends

- Specialization across the entire hardware/software stack is now the primary method for efficiency improvement
  - An exciting time to be in chip design!
- Ongoing battle between venerable spec-based design paradigms and newer agile-based paradigms
  - Guess which side I'm on :)
- Open-source design is beginning to gain some traction
- Industry is facing a big shortage of workers from design to manufacturing

More on these later

Application Trends



- **Machine learning and artificial intelligence (ML/AI)**
- Ultra-low-power sensors / wireless / IoT
- Cryptography / Privacy
- Health / biotechnology
- Graph mining
- Genomics
- Robotics